

REPORT ON PROJECT

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PROJECT NAME : Zudio Retail Sales Customer and Profitability Analysis.

PROJECT DOMAIN : Sales and Ecommerce

RAW DATASET LINK : [Zudio Sales test dataset](#)

CLEANED DATASET LINK :

[Zudio Retail Sales Customer and Profitability Analysis./Zudio SALES PROJECT 1 1.xlsx at main · Deepthi25B/Zudio Retail Sales Customer and Profitability Analysis.](#)

ZUDIO Retail sales Customer and Profitability Analysis.

Introduction:

The Zudio Retail Sales, Customer & Profitability Analysis project focuses on analyzing retail sales data to understand sales performance, profitability, product/category performance, and customer purchasing behavior. The project follows an end-to-end data analytics approach using Microsoft Excel, MySQL, Python, Matplotlib, Seaborn, and Power BI. The sales dataset was first cleaned and prepared in Excel, followed by data querying and aggregation in MySQL. Python was then used for exploratory analysis and RFM (Recency, Frequency, Monetary) customer segmentation, while Power BI was used to develop an interactive dashboard. This project is designed to transform raw retail transaction data into meaningful business insights that can support better sales, profitability, and customer-focused decisions.

Objectives

- To clean and prepare the Zudio retail sales dataset using Excel.
- To analyze sales revenue, profit, quantity, and order performance.
- To compare sales and profitability across categories, states, stores, and clothing types.
- To analyze monthly revenue and profit trends.
- To perform joins and aggregations using MySQL.
- To analyze customer purchasing behavior using RFM analysis in Python.
- To identify different customer segments based on recency, frequency, and monetary value.
- To create meaningful visualizations using Matplotlib and Seaborn.
- To develop an interactive Power BI dashboard for business decision-making.
- To identify important business insights and provide future recommendations.

Problem Statement:

Retail businesses generate large amounts of transactional data, but raw sales data alone does not provide clear information about business performance or customer behavior. The challenge was to clean the data, remove inconsistencies, handle missing values, and convert it into a usable format for meaningful visual insights.

Mission of the Project:

To transform Zudio retail transaction data into actionable business insights by analyzing sales performance, profitability, and customer behavior using modern data analytics tools.

Types of Analysis:

- **Descriptive Analysis:** Used to understand overall sales, revenue, profit, quantity, customers, and orders.
- **Sales Performance Analysis.**
- **Profitability Analysis.**
- **RFM Customer Segmentation.**
- **Exploratory Data Analysis:** Python was used to identify distributions, relationships, trends, and patterns using statistical summaries and visualizations.

DATA CLEANING IN DATA SET USING EXCEL

Convert given dataset into table

The entire dataset was first converted into a proper structured table/range. By selecting the range **ctrl+a and click insert tab** - select Table option.

Raw dataset:

A1		:	X	✓	f _x	State												
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	State	City	Category	Clothing T	Store Num	Store Type	Store Ope	Selling Are	Staff Coun	Order ID	Order Dat	Month	Customer	Product I	C Price	Quantity	Sales Profit	
2	Karnataka	Jalna	Women	Dresses	30	Owned	4/6/2022	950	17	88240	#####	November	233084	354	1958	5	3916	
3	Gujarat	Patna	Kids	Skirts	48	Owned	19-12-202	1200	6	78145	17-06-202	June	926472	741	2078	1	831.2	
4	Uttar Prad	Kolkata	Men	Hoodies	97	Rented	4/9/2023	1100	18	33363	17-08-202	August	853255	607	1445	2	1156	
5	West Beng	Kolkata	Women	Dresses	93	Owned	#####	1400	13	46682	27-03-202	March	106146	991	2469	6	5925.6	
6	West Beng	Guwahati	Men	Dresses	51	Owned	29-05-202	1400	19	58242	8/4/2024	April	264070	479	1105	6	2652	
7	West Beng	Nashik	Kids	Jackets	71	Owned	17-06-202	800	14	78800	25-04-202	April	326466	295	2206	4	3529.6	
8	Maharash	Lucknow	Kids	Jackets	44	Rented	29-04-202	800	6	43329	19-06-202	June	724817	222	2848	4	4556.8	
9	West Beng	Hyderabad	Women	Tops	74	Owned	16-11-202	950	18	23062	#####	June	870587	595	1694	8	5420.8	
10	Karnataka	Ahmedab:	Men	Shoes	71	Rented	23-11-202	1400	5	72223	24-01-202	January	790306	243	2024	1	809.6	
11	Karnataka	Hyderabad	Kids	Jeans	26	Rented	2/9/2021	1400	8	35882	2/2/2024	February	943541	910	2581	6	6194.4	
12	Maharash	Kolkata	Men	Hoodies	26	Rented	27-12-202	1300	9	13270	16-07-202	July	734952	639	1457	5	2914	
13	Uttar Prad	Nashik	Men	Jackets	90	Rented	#####	1000	11	84854	22-07-202	July	151454	878	2839	1	1135.6	
14	Maharash	Faridabad	Kids	Pants	5	Rented	#####	1200	3	97711	20-05-202	May	716554	740	2592	5	5184	
15	Gujarat	Noida	Kids	Skirts	74	Owned	#####	1200	19	30037	17-05-202	May	806957	255	642	8	2054.4	
16	Tamil Nad	Madurai	Kids	Sweaters	75	Owned	29-07-202	1000	5	97353	26-10-202	October	512228	963	2875	2	2300	

After Inserting Table:

PRODUCT NOTICE Excel hasn't been activated. To keep using Excel without interruption, activate before Friday, September 11, 2026. [Activate](#)

	A	B	C	D	E	F	G	H	I	J	K
	State	City	Category	Clothing Type	Store Number	Store Type	Store Open Date	Selling Area Size (sq ft)	Staff Count	Order ID	Order Date
1	Karnataka	Jalna	Women	Dresses	30	Owned	04-06-2022	950	17	88240	06-11-2024
2	Gujarat	Patna	Kids	Skirts	48	Owned	19-12-2020	1200	6	78145	17-06-2024
3	Uttar Pradesh	Kolkata	Men	Hoodies	97	Rented	04-09-2023	1100	18	33363	17-08-2024
4	West Bengal	Kolkata	Women	Dresses	93	Owned	11-04-2024	1400	13	46682	27-03-2024
5	West Bengal	Guwahati	Men	Dresses	51	Owned	29-05-2022	1400	19	58242	08-04-2024
6	West Bengal	Nashik	Kids	Jackets	71	Owned	17-06-2021	800	14	78800	25-04-2024
7	Maharashtra	Lucknow	Kids	Jackets	44	Rented	29-04-2021	800	6	43329	19-06-2024
8	West Bengal	Hyderabad	Women	Tops	74	Owned	16-11-2021	950	18	23062	11-06-2024
9	Karnataka	Ahmedabad	Men	Shoes	71	Rented	23-11-2023	1400	5	72223	24-01-2024
10	Karnataka	Hyderabad	Kids	Jeans	26	Rented	02-09-2021	1400	8	35882	02-02-2024
11	Maharashtra	Kolkata	Men	Hoodies	26	Rented	27-12-2022	1300	9	13270	16-07-2024
12	Uttar Pradesh	Nashik	Men	Jackets	90	Rented	10-12-2022	1000	11	84854	22-07-2024
13	Maharashtra	Faridabad	Kids	Pants	5	Rented	11-06-2020	1200	3	97711	17-05-2024
14	Gujarat	Noida	Kids	Skirts	74	Owned	03-10-2021	1200	19	30037	20-05-2024
15	Tamil Nadu	Madurai	Kids	Sweaters	75	Owned	29-07-2022	1000	5	97353	26-10-2024
16	Rajasthan	Noida	Women	Dresses	4	Rented	13-10-2020	1000	4	10924	19-03-2024
17	Maharashtra	Bhattacharya	Men	Pants	29	Owned	22-05-2020	950	19	89515	09-11-2024

Removed duplicates:

The dataset contained duplicate records based on key fields.

Inconsistent column:

- Checked the dataset structure.
- Removed unnecessary/blank columns.
- Checked missing values.
- Verified column names.
- Checked data types.
- The Date column contained inconsistent date formats, using both Slash and Hyphen format. Standardized all date entries by converting slashes (/) to hyphens (-).

Sales column:

Problem Identified:

The original dataset did not have a separate Sales Revenue column to directly calculate the total revenue generated from each sales transaction.

Action Taken:

A new **Sales Revenue** column was created by calculating the total sales value for each transaction using $\rightarrow \text{Price} \times \text{Quantity}$.

Steps Followed:

- Identify the Price and Quantity column to create New column
- Created a New Column named Sales Revenue.

- Identified the **Sales Revenue** column.
- Identified the **Sales Profit** column.
- Created the **Cost** column.

Original value		Standardized Column
Sales revenue	Sales Profit	Cost
₹ 9,790	₹ 3,916	₹ 5,874
₹ 2,078	₹ 831	₹ 1,247
₹ 2,890	₹ 1,156	₹ 1,734
₹ 14,814	₹ 5,926	₹ 8,888
₹ 6,630	₹ 2,652	₹ 3,978
₹ 8,824	₹ 3,530	₹ 5,294
₹ 11,392	₹ 4,557	₹ 6,835
₹ 13,552	₹ 5,421	₹ 8,131
₹ 2,024	₹ 810	₹ 1,214
₹ 15,486	₹ 6,194	₹ 9,292
₹ 7,285	₹ 2,914	₹ 4,371
₹ 2,839	₹ 1,136	₹ 1,703
₹ 12,960	₹ 5,184	₹ 7,776
₹ 5,136	₹ 2,054	₹ 3,082
₹ 5,750	₹ 2,300	₹ 3,450
₹ 23,376	₹ 9,350	₹ 14,026
₹ 16,696	₹ 6,678	₹ 10,018
₹ 6,897	₹ 2,759	₹ 4,138

Formula used:

→ Sales Revenue – Sales Profit

Revenue Category column:

The Sales Revenue values were numerical and varied considerably between transactions. It was difficult to quickly identify high-, medium-, and low-revenue transactions.

Action Taken

A new column named **Revenue Category** was created using the IF function.

Step Followed

- Created a new column named Revenue Category.
- Selected the Sales Revenue value for each row.
- Applied conditional logic using the IF function.

Original Value	Standardized Column
Sales revenue ▼	Revenue Catego ▼
₹ 9,790	Medium Revenue
₹ 2,078	Low Revenue
₹ 2,890	Low Revenue
₹ 14,814	High Revenue
₹ 6,630	Medium Revenue
₹ 8,824	Medium Revenue
₹ 11,392	High Revenue
₹ 13,552	High Revenue
₹ 2,024	Low Revenue
₹ 15,486	High Revenue
₹ 7,285	Medium Revenue
₹ 2,839	Low Revenue
₹ 12,960	High Revenue
₹ 5,136	Medium Revenue
₹ 5,750	Medium Revenue
₹ 23,376	High Revenue
₹ 16,696	High Revenue
₹ 6,897	Medium Revenue

Formula Used:

→ =IF(Q2>=10000,"High Revenue", IF(Q2>=5000,"Medium Revenue", "Low Revenue "))

Profit Status Column:

Problem Identified

Although the dataset contained Sales Profit, it was difficult to quickly identify which transactions generated high profit and which generated relatively lower profit.

Action Taken

A new column named Profit Status was created.

Steps Followed

- Created a new column named Profit Status.
- Selected the Sales Profit value for each row.
- Applied an IF condition.

Original value	Standardized Column

Sales Profit ▾		Profit Status ▾
₹ 3,916		Low Profit
₹ 831		Low Profit
₹ 1,156		Low Profit
₹ 5,926		Low Profit
₹ 2,652		High Profit
₹ 3,530		Low Profit
₹ 4,557		Low Profit
₹ 5,421		High Profit
₹ 810		High Profit
₹ 6,194		High Profit
₹ 2,914		Low Profit
₹ 1,136		High Profit
₹ 5,184		Low Profit
₹ 2,054		Low Profit
₹ 2,300		Low Profit
₹ 9,350		High Profit

Formula Used:

→ **=IF(S2>=4000,"High Profit", "Low Profit")**

XLOOKUP:

- **= XLOOKUP(A2,'Zudio_sales_data-selected-column'! E:E,'Zudio_sales_data-selected-column'!A:A)**
- Using XLookup created a separate sheet called Store master.

Store Number	State	Store type	Staff count					
30	Karnataka	Owned	17					
48	Gujarat	Owned	6					
97	Uttar Pradesh	Rented	18					
93	West Bengal	Owned	13					
51	West Bengal	Owned	19					
71	West Bengal	Owned	14					
44	Maharashtra	Rented	6					
74	West Bengal	Owned	18					
71	West Bengal	Owned	14					
26	Karnataka	Rented	8					
26	Karnataka	Rented	8					
90	Uttar Pradesh	Rented	11					
5	Maharashtra	Rented	3					
74	West Bengal	Owned	18					
75	Tamil Nadu	Owned	5					
4	Rajasthan	Rented	4					
29	Maharashtra	Owned	19					
43	Rajasthan	Owned	8					
►	Zudio_sales_data-selected-column	Store_master		Pivot table 1	Pivot table 2	Pivot table 3		

JOINS AND AGGREGATION USING SQL

- After preparing the data, the cleaned dataset was imported into MySQL.
- Additional summary tables were created for analytical purposes:
 - ✓ order summary
 - ✓ customer summary
 - ✓ product summary

Customer Summary:

```
2    --- customer summary
3 • CREATE TABLE customer_summary AS
4    SELECT
5        `Customer ID`,
6        COUNT(DISTINCT `Order ID`) AS Total_Orders,
7        SUM(`Sales revenue`) AS Total_Revenue,
8        SUM(`Sales Profit`) AS Total_Profit,
9        SUM(Quantity) AS Total_Quantity
0    FROM zudio_sales
1    GROUP BY `Customer ID`;
```

Order Summary

```
CREATE TABLE order_summary AS
SELECT
    `Order ID`,
    MIN(`Order Date`) AS Order_Date,
    MIN(`Customer ID`) AS Customer_ID,
    SUM(Quantity) AS Total_Quantity,
    SUM(`Sales revenue`) AS Order_Revenue,
    SUM(`Sales Profit`) AS Order_Profit
FROM zudio_sales
GROUP BY `Order ID`;
SHOW tables;
```

Joins :

---- JOINS AND AGGREGATION

- SELECT

```
c.`Customer ID`,
c.Total_Orders,
SUM(o.Order_Revenue) AS Total_Order_Revenue,
SUM(o.Order_Profit) AS Total_Order_Profit
FROM customer_summary c
INNER JOIN order_summary o
    ON c.`Customer ID` = o.Customer_ID
GROUP BY
```

Using Foreign and Primary key created a ERR diagram using Customer Id and Order Id.

```
ALTER TABLE order_summary
ADD PRIMARY KEY (`Order ID`);

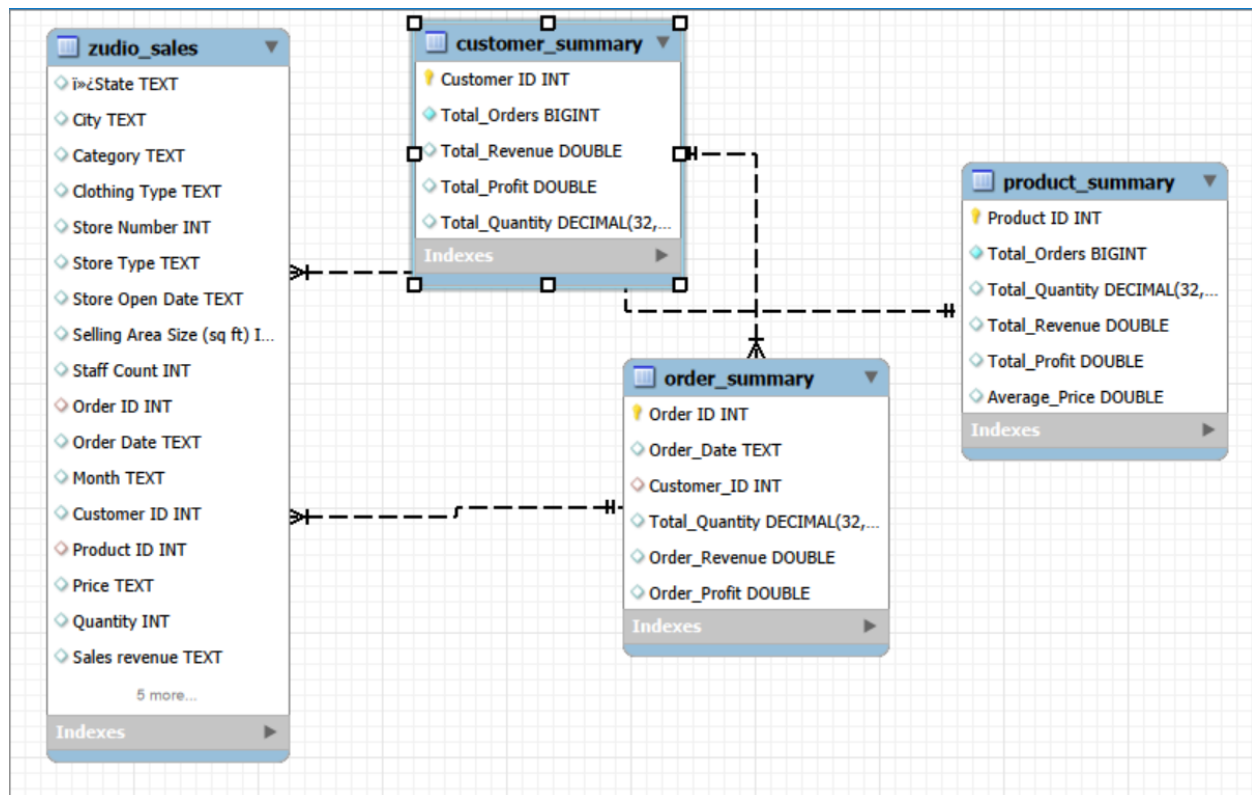
ALTER TABLE zudio_sales
ADD CONSTRAINT fk_sales_order
FOREIGN KEY (`Order ID`)
REFERENCES order_summary(`Order ID`);

ALTER TABLE customer_summary
ADD PRIMARY KEY (`Customer ID`);

ALTER TABLE order_summary
ADD CONSTRAINT fk_order_customer
FOREIGN KEY (`Customer_ID`)
REFERENCES customer_summary(`Customer ID`);

ALTER TABLE product_summary
```

ERR Diagram:



RFM Analysis Using Python

Python was used for customer analysis using Pandas and NumPy.

RFM stands for:

Recency

- ✓ How recently a customer made a purchase.

Frequency

- ✓ How frequently a customer made purchases.

Monetary

- ✓ How much money a customer spent.

The RFM analysis produced 7,870 unique customers.

```
# RFM Analysis ----> R - Recency, F - Frequency, M - Monetary.  
  
analysis_date = df['Order Date'].max() #----> set analysis using date  
  
print("Analysis Date:", analysis_date)
```

Analysis Date: 2024-12-11 00:00:00

```
rfm = df.groupby('Customer ID').agg(      # ----> RFM table  
    Recency=('Order Date',  
            lambda x: (analysis_date - x.max()).days),  
  
    Frequency=('Order ID', 'nunique'),  
  
    Monetary=('Sales revenue', 'sum')  
) .reset_index()  
  
rfm.head()
```

RFM score:

```

## RFM Score.
rfm['R_Score'] = pd.qcut( # ---> Recency
    rfm['Recency'].rank(method='first'),
    5,
    labels=[5,4,3,2,1]
).astype(int)

rfm['F_Score'] = np.where( #---> Frequency
    rfm['Frequency'] >= 2,
    5,
    1
)

rfm['M_Score'] = pd.qcut( #----> Monetary
    rfm['Monetary'].rank(method='first'),
    5,
    labels=[1,2,3,4,5]
).astype(int)

```

Customer Segmentation:

```

## Customer Segmentation.
def segment_customer(row):

    r = row['R_Score']
    f = row['F_Score']
    m = row['M_Score']

    if r >= 4 and f >= 4 and m >= 4:
        return 'Champions'

    elif r >= 3 and f >= 4:
        return 'Loyal Customers'

    elif r >= 4 and f <= 2:
        return 'New Customers'

    elif r <= 2 and m >= 4:
        return 'At Risk'

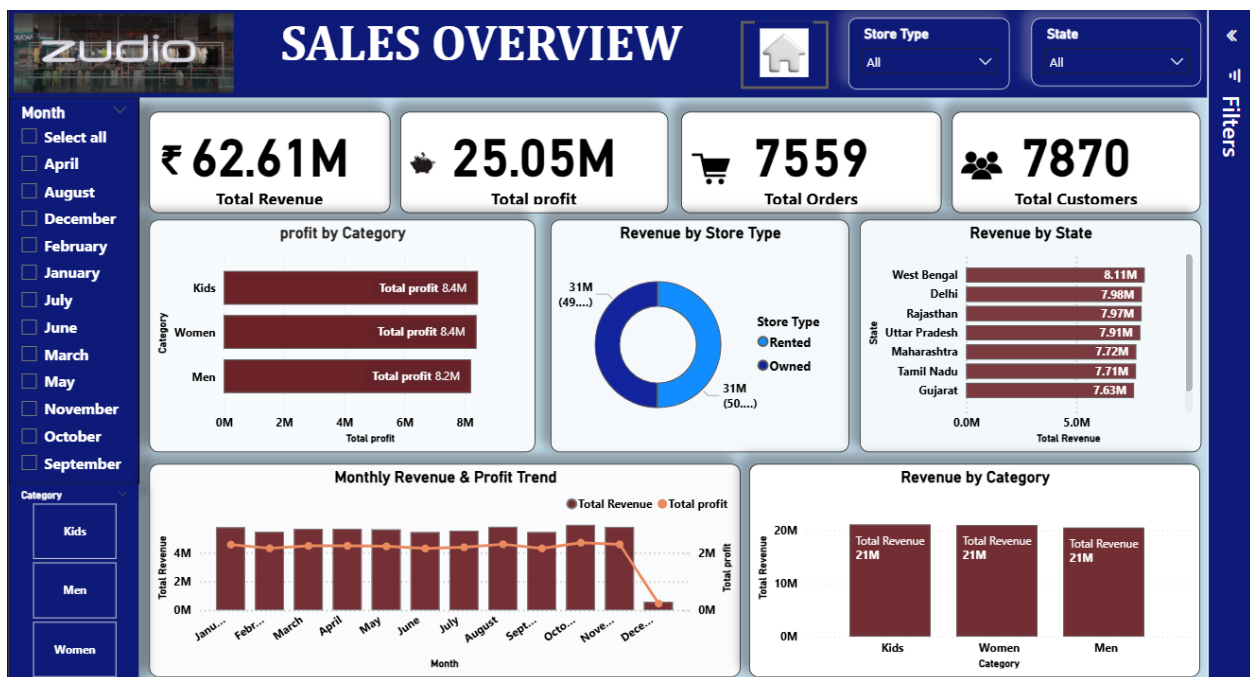
```

Dashboard:

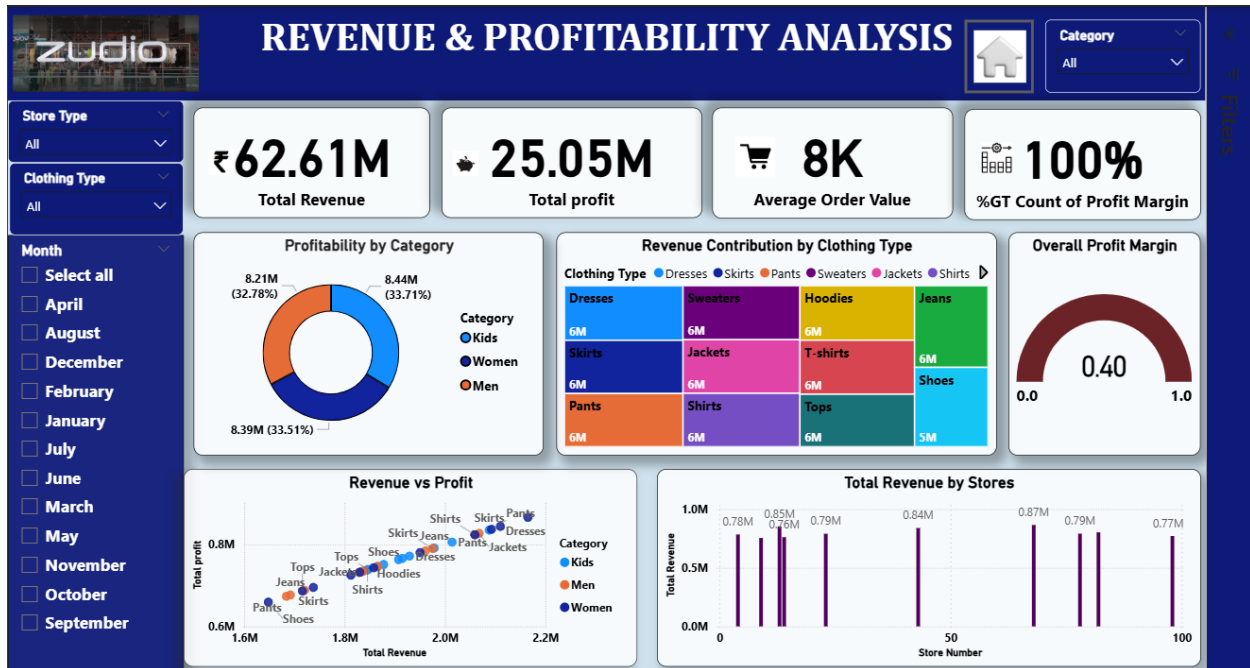
Home page



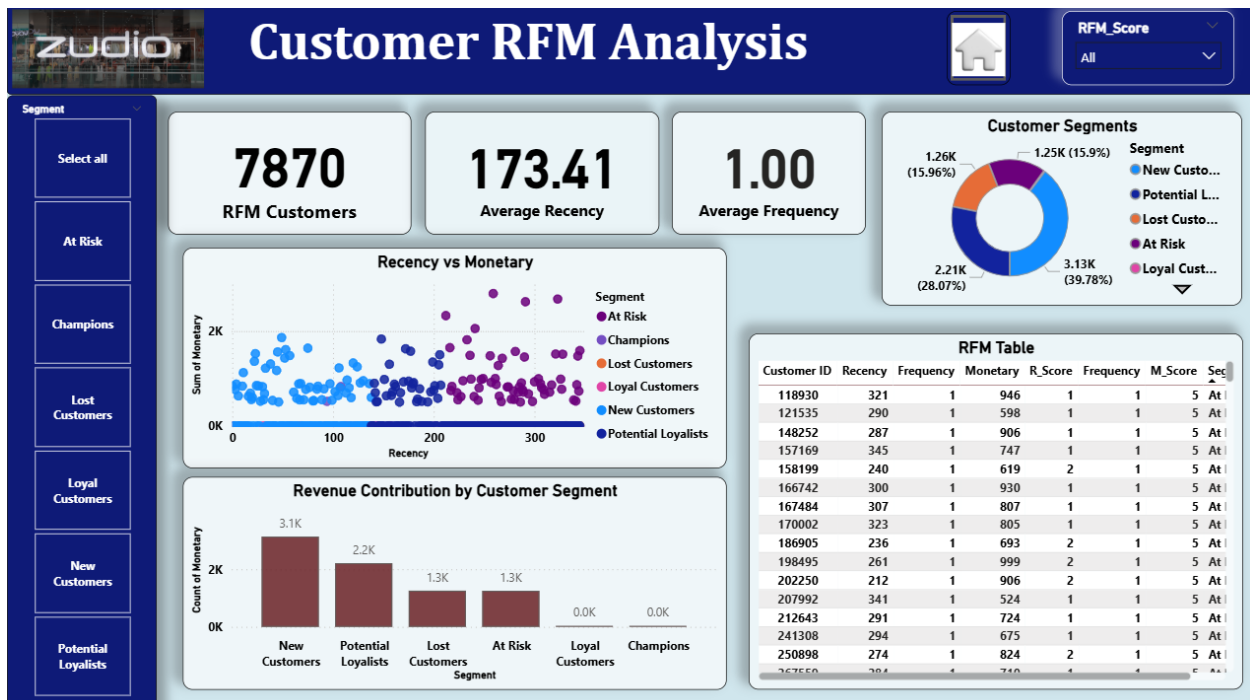
Sales Overview



Revenue and Profitability Analysis



Customer RFM Analysis



Key Insights:

- The analysis produced several important findings.

1. Overall Business Performance

- The dataset generated approximately:
- Total Revenue: ₹62.61 Million
- Total Profit: ₹25.05 Million
- Total Quantity Sold: 35,699
- Total Orders: 7,559
- Total Customers: 7,870

The overall profit margin is approximately 40%.

- Category Performance: The Kids category generated the highest revenue and profit.

- State Performance → The highest revenue-generating states included:

- West Bengal
- Delhi
- Rajasthan
- Uttar Pradesh
- Maharashtra

These locations can be studied further to understand what factors contribute to their stronger sales performance.

- Store Type → Owned and rented stores both contribute significantly to revenue and profit. So the analysis shows that rented stores have slightly higher total profit than owned stores in this dataset.
- Customer Repeat Purchase → The most important customer insight is the low repeat-purchase rate. Out of 7,870 customers, 7,841 customers made only one purchase, while only 29 customers made two purchases. This indicates a major opportunity to improve customer retention and repeat purchases.
- RFM Segmentation → The largest customer segment is New Customers, followed by Potential Loyalists. There are relatively few Champions and Loyal Customers, which suggests that converting new and potential customers into repeat customers could be an important business opportunity.

- At-Risk and Lost Customers→ The RFM analysis identifies a significant number of customers as At Risk and Lost Customers. These customers could be targeted with personalized offers, loyalty benefits and re-engagement campaigns.

Future Recommendation:

- Increase Repeat Purchases.
- Target New Customers.
- Re-engage At-Risk Customers.
- Recover Lost Customers.
- Focus on High-Performing Categories.
- Monitor Store Performance.
- Use Dashboard-Based Decision Making.

Conclusion:

The Zudio Retail Sales, Customer & Profitability Analysis project used Excel, MySQL, Python, and Power BI to analyze sales, profitability, and customer behavior. The analysis generated ₹62.61 million revenue and ₹25.05 million profit, with the Kids category performing best. RFM analysis highlighted low repeat purchases and significant New, Potential Loyalist, At Risk, and Lost customer segments. Overall, the project provides actionable insights for improving customer retention, repeat purchases, and business performance.

